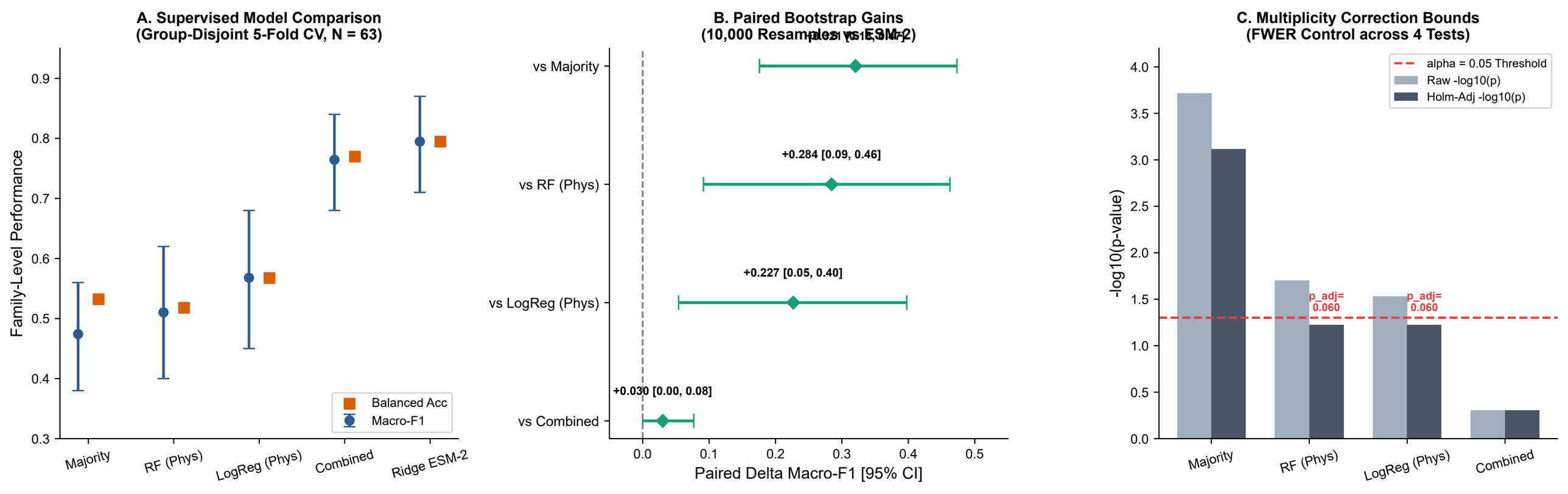

A. Supervised Model Comparison
(Group-Disjoint 5-Fold CV, N = 63)
Family-Level Performance
0.9
0.8
0.7
0.6
0.5
0.4
0.3
Balanced Acc
Macro-F1
Majority
RF (Phys)
LogReg (Phys)
Combined
Ridge ESM-2
B. Paired Bootstrap Gains
(10,000 Resamples vs ESM-2)
vs Majority
vs RF (Phys)
vs LogReg (Phys)
vs Combined
+0.284 [0.09, 0.46]
+0.227 [0.05, 0.40]
+0.030 [0.00, 0.08]
0.0
0.1
0.2
0.3
0.4
0.5
Paired Delta Macro-F1 [95% CI]
C. Multiplicity Correction Bounds
(FWER Control across 4 Tests)
alpha = 0.05 Threshold
Raw -log10(p)
Holm-Adj -log10(p)
-log10(p-value)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
p_adj=
0.060
p_adj=
0.060
Majority
RF (Phys)
LogReg (Phys)
Combined